

derived from the subject set-up.

Given a connected singular stratum and a tubular neighbourhood $\text{Tub}_{3\epsilon}(S) \subset X$, we can expand a Hermitian Dirac bundle structure $(E, \text{cl}_g, h, \nabla^h)$ on (X, g) with respect to the Riemannian distance to the singular stratum S and identify the zeroth order term as a (singular) conically fibred (CF) Hermitian Dirac bundle structure on the normal cone bundle $\nu_0 : X_0 = NS/\text{Isot}(S) \rightarrow S$ of S .

A **Gromov-Hausdorff resolution of a Riemannian orbifold** (X, g) is given by a smooth family of (singular) Riemannian orbifolds (X^t, g^t) and morphisms

$$\rho^t : X^t \dashrightarrow X$$

such that ρ^t restricts to an orbifold diffeomorphism $(\rho^t)^{-1}(X^{\text{reg}}) \cong X^{\text{reg}}$, the exceptional set $\Upsilon^t = (\rho^t)^{-1}(X^{\text{sing}})$ is of codimension ≥ 2 and

$$(X^t, g^t) \xrightarrow[\text{GH}]{t \rightarrow 0} (X, g) \quad \text{and} \quad (X^t \setminus \Upsilon^t, g^t) \xrightarrow[\text{C}^\infty]{t \rightarrow 0} (X^{\text{reg}}, g)$$

In particular $\text{vol}_{g^t}(\Upsilon^t) \xrightarrow[t \rightarrow 0]{} 0$ and $\rho_*^t g^t \xrightarrow[\text{C}^\infty]{t \rightarrow 0} g$.

Such resolutions are usually constructed via resolutions of the CF-Hermitian Dirac bundle over the normal cone bundle consist of a family of **asymptotically conically fibred (ACF)** Hermitian Dirac bundle-structures on a Riemannian fibration subject to a list of assumptions. The resolutions (X^t, g^t) and the resolutions of the Hermitian Dirac bundles are then constructed by interpolating between the ACF structures on the resolutions $(X_{t^2 \cdot \zeta}, g_{t^2 \cdot \zeta})$ of the normal cone (X_0, g_0) and the Hermitian orbifold Dirac bundle structures on X along a (shrinking) tubular neighbourhood of S .

In order to analyse the resulting family of Dirac operators D^t on the resolved orbifolds (X^t, g^t) we split the latter into three topological regions: the ACF part, the CF-part and the CSF-part. We will proceed by analysing all the different components individually in order to derive a uniform elliptic theory on the total space of the resolution.

We begin by considering the CF-part of the orbifold resolution. This part more generally referred to as the neck part of the gluing construction, can be realised as an annulus subbundle of the normal cone bundle of shrinking outer diameter. In this region, the **analytic behaviour of D^t is determined by the normal operator $\widehat{D}_0$ of D** , which can be seen as the leading order operator in an asymptotically expansion of D close to the singular stratum. By using essentially an elaborated version of separation of variables we can explicitly expand the kernel of $\widehat{D}_0$. Using this expansion, we argue that $\widehat{D}_0$, D and $\widehat{D}_{t^2 \cdot \zeta}$, the latter being a resolution of $\widehat{D}_0$, should be realised as bounded operators on **weighted function spaces**, subject to **boundary conditions**. By carefully choosing the right function spaces the normal operator can be realised as an isomorphism of Banach spaces; $\widehat{D}_{t^2 \cdot \zeta}$ and D as Fredholm maps.

We continue by defining the so called approximate (co)kernel of D^t by interpolating the (co)kernel of the operators $\widehat{D}_{t^2 \cdot \zeta}$ and D . One of the scopes of this paper is to prove the exactness of linear gluing, by virtue of the exactness of the sequence

$$\ker(D^t) \xhookrightarrow{i^t} \text{xker}(D^t) \xrightarrow{\text{ob}^t} \text{xcoker}(D^t) \xrightarrow[p^t]{\dashrightarrow} \text{coker}(D^t).$$

In order to prove the above, we will need to study the so called (anti-) gluing equations in the style of Hutchings-Taubes obstruction bundle gluing [HT07, HT09].

Further, we will prove the existence of right-inverses to the operator D^t and construct adapted norms, for which these right-inverse are uniformly bounded. This problem arises due to the **collapsing**